

Questioning Representational Optimism in Deep Learning: The Fractured Entangled Representation Hypothesis

Alexandra Karali, Magdalini Maria Pliatsika, Georgios Kouseris,
Georgios Oikonomoy, Stylianos Vantarakis

Abstract

Recent work has challenged the prevailing “representational optimism” in deep learning, proposing the Fractured Entangled Representation (FER) hypothesis: the idea that models trained via conventional Stochastic Gradient Descent (SGD) often achieve high performance through disorganized, heuristic-based internal structures rather than the Unified Factored Representations (UFR) found in open-ended evolutionary systems. This paper empirically investigates the FER hypothesis by contrasting SGD-trained Compositional Pattern Producing Networks (CPPNs) with those evolved via Picbreeder.

We extend the original analysis through a series of novel experiments targeting the mechanisms of representational formation. First, we demonstrate that modifying the loss function with orthogonality and sparsity regularization significantly enhances the interpretability of the latent space compared to baseline SGD. Second, by analyzing evolutionary lineages, we show that open-ended search organizes variation around low-dimensional, structurally meaningful axes, such as symmetry, which function as continuous, tunable properties. Conversely, our layer-wise analysis reveals that while regularities like rotational symmetry are fragile in early layers, they are consolidated into stable abstractions in deeper layers. Finally, we investigate the “scaling” argument by training CPPNs on larger datasets of conditioned images. We find that while scaling introduces mild pressure toward structure and representational stability, it fails to resolve FER, suggesting that data scale alone is insufficient to induce unified internal understanding.

1 Introduction

This paper challenges “representational optimism”—the belief that scaling data and compute naturally leads to good internal representations. The authors argue that while scaling improves downstream performance, it does not necessarily create coherent internal models of the world. By comparing networks evolved through open-ended search to those trained via conventional Stochastic Gradient Descent (SGD), the paper reveals that models with identical outputs can have radically different underlying structures.

1.1 Background

To investigate internal representations, the authors use a “toy model” setup that allows for complete transparency, unlike modern “black box” Large Language Models (LLMs).

1.1.1 CPPNs (Compositional Pattern Producing Networks)

These are neural networks that take pixel coordinates (x, y) as input and output color values. Because they map a 2D space, researchers can visualize the behavior of every single hidden neuron as an image, revealing exactly how the network constructs its output.

1.1.2 Picbreeder

This was a platform where humans “bred” images via an open-ended evolutionary process rather than training for a specific target. This process produced complex images, such as a symmetrical skull, which possessed elegant, modular internal structures.

1.2 FER & UFR

The study demonstrates that when a fresh network is trained via SGD to perfectly replicate a Picbreeder image, it succeeds at the output level but fails internally.

Fractured Entangled Representation (FER) This is the disorganized state found in SGD-trained networks. Information regarding a single concept (like symmetry) is split into disconnected pieces (“fractured”) and inappropriately mixed with other functions (“entangled”). It is analogous to “spaghetti code”—messy, redundant, and brittle.

Unified Factored Representation (UFR) This is the organized state found in evolved Picbreeder networks. Concepts are distinct, modular, and reused effectively, similar to clean software design, which supports better adaptation.

1.3 Imposter Intelligence

The authors coin the term “Imposter Intelligence” for models that appear flawless on the surface (output) but rely on FER underneath. For example, an SGD-trained model can draw a perfect skull but lacks the internal concept of “symmetry,” meaning it cannot generalize or adapt creatively. The paper suggests that modern foundation models may be similar imposters: passing benchmarks through massive memorization and heuristics rather than true, unified understanding.

2 Our Experiments

2.1 Loss Function Regularizations

To promote the learning of disentangled representations, we modified the model’s objective function by incorporating orthogonality regularization and activation sparsity regularization. The primary objective of orthogonality regularization is the decorrelation of feature maps. This is achieved by computing the inner product between all pairs of feature maps to construct a correlation matrix. The regularization term penalizes the divergence of this matrix from the identity matrix I , where a value of 1 on the diagonal and 0 elsewhere indicates that each feature map is statistically independent of the others. Theoretically, this constraint encourages the model to develop mutually exclusive internal representations. Complementarily, we implemented a sparsity constraint by imposing a penalty based on the L_1 norm (absolute value) of the activations. This encourages the emergence of sparse neural networks, where specific neurons are selectively activated only upon the detection of distinct, localized patterns.

2.1.1

Experimental Results and Analysis Our empirical observations confirm that these modifications to the loss function significantly enhance the interpretability of the model’s internal representations compared to standard Stochastic Gradient Descent (SGD). By performing weight sweeps, we can more easily map latent variables to visual outputs. As illustrated in the first figure, adjusting the weights directly influences the spatial distribution of high-intensity (white) regions. In the second figure, weight perturbations correspond to semantic changes, such as

the inter-ocular distance. While extracting these insights requires meticulous observation, the latent space exhibits a much clearer structure than the baseline model.

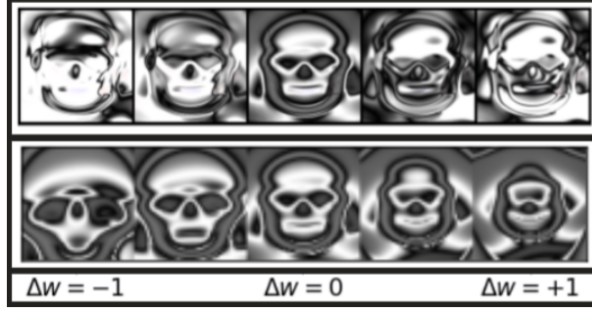


Figure 1: Visualized Results

2.2 Transfer Learning and Feature Complexity

We further investigated whether a model could derive complex, diverse features by utilizing the foundational characteristics learned by a Compositional Pattern-Producing Network (CPPN). To test this, we utilized a pre-trained CPPN, freezing the initial 80% of the layers and performing SGD-based optimization only on the remaining 20% of the architecture.

The results indicate a successful transition to higher-level complexity. For instance:

- In the top picture, a specific weight parameter now controls the inclination and positioning of the jaw.
- In bottom picture, adjustments to a single weight regulate the proportions and symmetry of the eyes.



Figure 2: Visualized Results

2.3 Low-Dimensional Structure in CPPN Evolution

In the experiment we investigated whether an open-ended evolutionary process develops regularities i.e., systematic, reusable structural patterns such as symmetry, along a concrete evolutionary trajectory in Picbreeder. To do so, we selected a single evolutionary lineage and collected a sequence of successive genomes spanning its evolutionary history. Each genome was originally encoded as a NEAT graph. Following the layerization procedure described in the paper, we converted the original NEAT graph into a functionally equivalent, fixed-depth, fixed-width, layer-based network. This transformation preserves the exact computation of the evolved CPPN while embedding it into a common parameterization suitable for systematic analysis. This step preserves the computation performed by each genome while yielding a fixed-length parameter vector, allowing all lineage members to be compared directly in the same weight space.

We then applied Principal Component Analysis (PCA) to the resulting set of parameter vectors to identify dominant directions of variation across the lineage. PCA provides a low-dimensional coordinate system that summarizes how the lineage moves through parameter space over evolutionary time. To interpret these directions in terms of regularities in the phenotype, we performed controlled traversals along the leading principal axis. Specifically, we generated images by shifting the parameter vector in both directions along the first component and rendering the corresponding CPPN outputs. This produced a sequence of phenotypes that reveal which morphological properties are systematically modulated by that axis.

Because our focus was explicitly on regularities, we quantified mirror symmetry as a canonical regularity that is both visually salient and computationally measurable. For each generated image along the axis, we computed a symmetry score by comparing the image to its mirror-reflected version, enabling us to test whether symmetry changes smoothly and consistently as a function of position along the dominant evolutionary direction. In this way, PCA was not used merely as a descriptive dimensionality reduction tool, but as a means to probe whether evolution organizes variation around interpretable regularities in the generated structures.

2.3.1 Results

The first principal component explained approximately 84% of the total variance in the lineage parameter vectors, indicating that the evolutionary trajectory is effectively low-dimensional despite the high dimensionality of the underlying network representation. Traversing this axis produced smooth, coherent morphological transitions, suggesting that the axis corresponds to a structured direction in phenotype space rather than capturing arbitrary parameter drift. Crucially, the symmetry analysis revealed a monotonic relationship between position along the first principal component and the degree of mirror symmetry in the generated images: symmetry increased in one direction along the axis and decreased in the other (Figure 3). The genomes observed in the actual lineage clustered around an intermediate region of this continuum, rather than at an extreme of maximal symmetry. These results support the view that evolution in this lineage is organized around a dominant structural direction that modulates a fundamental regularity of the phenotype. Symmetry behaves as a continuous and tunable property within the representation, rather than a fixed constraint that is either present or absent. At the same time, the lineage does not converge to maximal regularity; instead, it stabilizes at a balanced regime where regular structure is maintained while allowing additional variation and differentiation. Overall, the findings suggest that a seemingly complex open-ended evolutionary process can be structured around a simple, low-dimensional axis that controls key regularities of form.

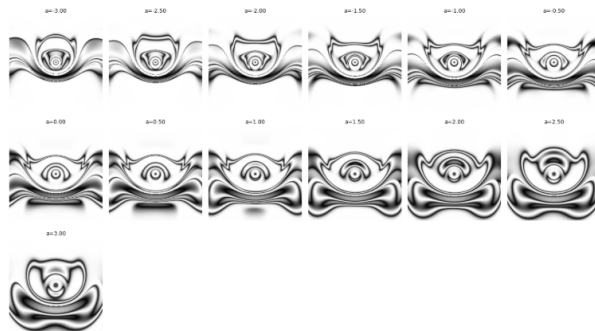


Figure 3: Traversal along the dominant principal axis.

2.4 Symmetry-Preserving Directions in Weight Space

Now, we inverted the logic of the first analysis. Instead of asking whether the dominant axes in parameter space preserve symmetry, we asked whether directions that preserve symmetry are

themselves structurally important. In other words, rather than testing if significant axes maintain regularity, we investigated whether regularity-preserving directions are significant within weight space.

To this end, we searched for directions in the network’s weight space that modify the generated image in a controlled manner while maintaining its fundamental mirror symmetry along the y-axis. We began by applying a large number of small random perturbations to the network’s weights. Each perturbation defined a direction in parameter space. These directions were evaluated according to two criteria: (1) the magnitude of visual change induced in the generated image, and (2) the degree to which mirror symmetry was preserved.

We retained only those perturbation directions that produced substantial perceptual variation without disrupting the global symmetric structure of the image. These filtered directions represent candidate axes along which meaningful transformation occurs without structural breakdown. We then performed dimensionality analysis within this subset of “good” directions to identify dominant axes in weight space. Finally, we conducted a weight sweep along one of these axes to characterize its functional effect on the generated image.

2.4.1 Results

Sweeping along the identified axis revealed a consistent and structured pattern of change (Figure 4). As the scalar coefficient α increased in the positive direction, the generated image became progressively more abstract, with reduced contrast and softer, less sharply defined edges. Conversely, for negative values of α , edges became sharper and contours more pronounced, producing a clearer and more defined structure.

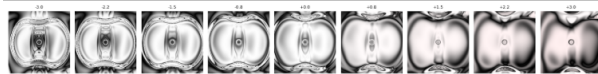


Figure 4: Weight sweep along a symmetry-preserving axis.

Importantly, throughout this traversal, the global mirror symmetry of the image remained intact. The transformations did not introduce chaotic distortions or asymmetric artifacts; instead, they modulated visual properties in a smooth and coherent manner. These results indicate that the identified axis encodes a stable and non-random direction of variation in weight space. It does not correspond to noise or arbitrary perturbation but rather to a structured degree of freedom that regulates a global morphological property specifically, the sharpness and contrast of the image while preserving its underlying symmetry. Thus, the experiment demonstrates that symmetry-preserving directions in weight space can be both meaningful and structurally dominant. The network appears to contain organized subspaces in which controlled variation of perceptual attributes is possible without destroying core regularities. This finding further supports the view that the representation is not purely entangled or chaotic, but contains coherent axes governing interpretable transformations.

2.5 Emergence of Rotational Symmetry Across Layers

In the experiment, we examined how circular (rotational) symmetry is internally represented within a CPPN. We began with an open-ended CPPN from Picbreeder, evolved through NEAT, which exhibits a clear circular structure in its final output.

To enable neuron-level analysis, the original NEAT graph was converted into an equivalent layer-based CPPN via a layerization procedure. This transformation preserved the exact functional behavior of the network while restructuring it into a standard layered architecture, allowing systematic inspection of individual neurons and their activation maps.

We then computed a quantitative measure of circular symmetry (rotation invariance) for the activation maps of all neurons in the network. For each neuron, its activation over the entire input grid was treated as an image, and its rotational consistency was evaluated. Trivial cases (e.g., constant or near-constant activations) were filtered out to avoid artificially inflated symmetry scores. Based on this measure, we identified neurons exhibiting high degrees of circular symmetry.

Subsequently, we performed activation sweeps on these selected neurons. Specifically, we modulated their activation values while keeping all weights fixed, allowing us to causally assess how each neuron contributes to the global circular structure of the generated image.

2.5.1 Results

The results reveal a clear layer-dependent organization of circular symmetry within the network. Activation sweeps in layer 3 (early layer) reveal that circular symmetry is not yet robustly encoded at early stages of computation (Figure 5). Modulating the selected rotation-invariant neuron leads to substantial distortions of the global structure, particularly for negative values of α , where the circular form becomes fragmented or partially disrupted. Although traces of radial organization remain visible, the symmetry is fragile and easily destabilized by perturbation. This suggests that, in early layers, rotational regularity is not yet consolidated as a stable representational factor but instead depends on delicate interactions across distributed features.

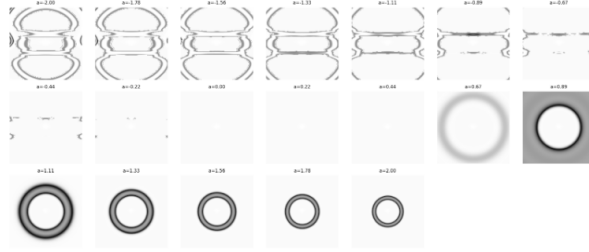


Figure 5: Activation sweep in layer 3

In layer 8 (intermediate layer), activation sweeps exhibit a qualitatively different behavior compared to early layers. Circular structure is preserved consistently across a wider range of α , while the modulation induces larger and more structured geometric changes. In particular, varying α produces a systematic change in the radius of the circular pattern, effectively expanding or contracting the ring in a smooth and radially coherent manner (Figure 6). Unlike early layers, where perturbations tend to destabilize symmetry, here the transformations remain globally organized and symmetric. The neuron no longer appears to construct symmetry from low-level components; instead, it modulates an already established circular structure by controlling its scale and intensity. This intermediate stage reflects a transition from fragile encoding toward a progressively stabilized and more geometrically meaningful representation.

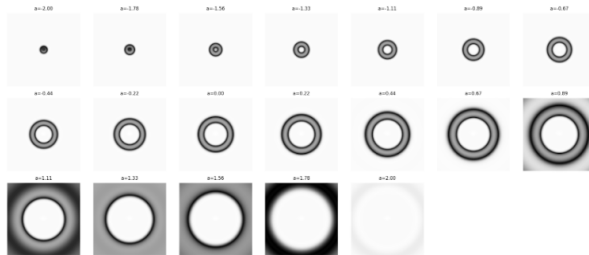


Figure 6: Activation sweep in layer 8

In layer 15 (late layer), activation sweeps demonstrate a fully stabilized encoding of rotational symmetry (Figure 7). Across the entire range of α , global circular structure remains intact, and modulation primarily affects contrast, sharpness, or radial intensity rather than geometric organization. The transformation is smooth and globally coherent, without introducing asymmetries or structural breakdown. At this depth, circular symmetry functions as a causally controllable and structurally distinct representational factor, indicating that the network has consolidated rotational regularity into a stable higher-level abstraction.

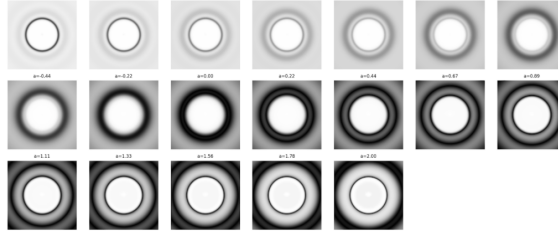


Figure 7: Activation sweep in layer 15

Overall, the findings show that circular symmetry is not uniformly distributed across the network. Rather than being globally encoded from the outset, it gradually emerges through the layered computation and ultimately becomes represented as a functionally distinct and causally meaningful factor. The symmetry is therefore not an incidental byproduct of low-level features, but an organized structural property that crystallizes in higher layers of the network.

2.6 Dataset Generation

The objective of this experimental study was to examine whether training a Compositional Pattern Producing Network (CPPN) with conventional stochastic gradient descent (SGD) on a dataset of structurally similar images can induce more structured internal representations, potentially moving the network away from Fractured Entangled Representations (FER) and toward more Unified Factored Representations (UFR), as discussed in the FER hypothesis. To investigate this question, we constructed an initial dataset of 60 similar images, as shown in figure 8, using the open-ended CPPN model described in the initial paper. We used the weight IDs identified in the initial paper as semantically meaningful and varied them in pairs within a limited range, generating controlled variations while preserving the base skull structure.



Figure 8: A subset of the generated dataset.

To enable the CPPN to model multiple images, the architecture was modified to accept an additional input vector z , corresponding to the identity of each image. The network therefore learned a conditional mapping from spatial coordinates and image identity to pixel color values, effectively learning $(x, y, d, z) \rightarrow (h, s, v)$. The vector z allowed the model to selectively generate

any image from the dataset. Training was performed using mean squared error across all pixels and all images jointly. After successful optimization, we analyzed the internal neuron activations by visualizing each neuron’s output over the full spatial grid, enabling direct inspection of the internal representational structure. We further conducted systematic weight sweeps to evaluate the semantic stability and locality of individual weight perturbations. To explore the effect of dataset scaling, the dataset was then expanded to 120 images and the same conditional architecture and training procedure were applied.

2.6.1 Results

The network successfully learned to reconstruct all images in both the 60-image and 120-image datasets with low reconstruction error, as shown in figure 9



Figure 9: Partial results of the CPPN trained with SGD on a dataset of similar images.

However, examination of the internal representations revealed that increasing the dataset size did not eliminate the entangled structure characteristic of FER. Most neurons exhibited complex, visually chaotic and highly entangled patterns without clear modular decomposition. However, a partial sharing could be identified since the differences between images were primarily expressed in higher layers, while earlier layers largely encoded a shared base structure resembling an “average” skull. This suggests that the model learned a common foundational representation and subsequently modulated it through the conditioning vector z . It has to be noted, though, that unlike single-image training, we observed the emergence of individual weight IDs with relatively localized and stable effects. Certain weights were found to control specific semantic properties, such as eye size, mouth opening, brightness, and global scaling, without causing complete structural collapse. Although these effects were not perfectly disentangled and still exhibited some cross-feature interference, they were clearly more stable than the single-image FER case, where almost any small change causes the image to break down chaotically.

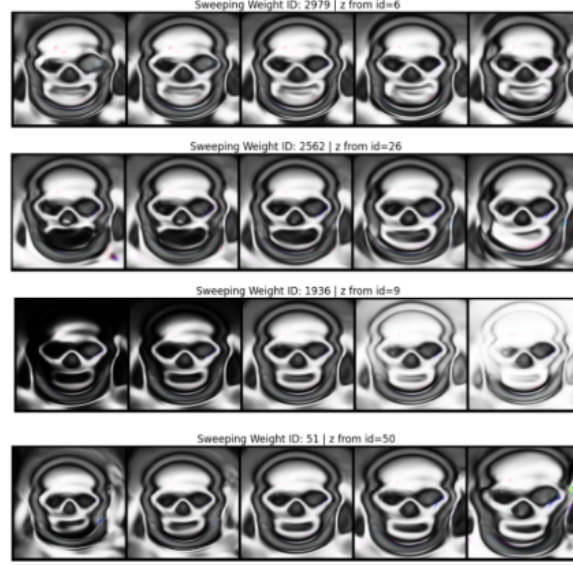


Figure 10: Weight sweeps of selected weight IDs in the SGD-trained CPPN, showing controlled variations across conditioned images without severe structural degradation.

Another important finding was that some weight IDs exhibited consistent semantic effects across multiple images. For example, as shown in figure 11, weight ID 1909 consistently modulated brightness across different conditioned outputs, while largely preserving other structural attributes such as jaw shape and mouth configuration. This cross-image consistency indicates partial reuse of internal structure and suggests the presence of limited representational stabilization.

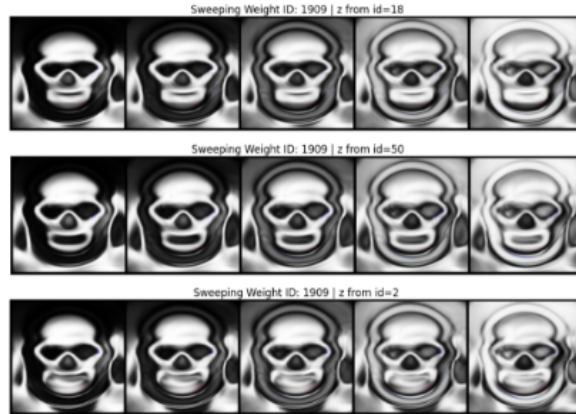


Figure 11: Sweeping weight ID 1909 across three images, showing that it consistently changes brightness while keeping the unique features of each image intact.

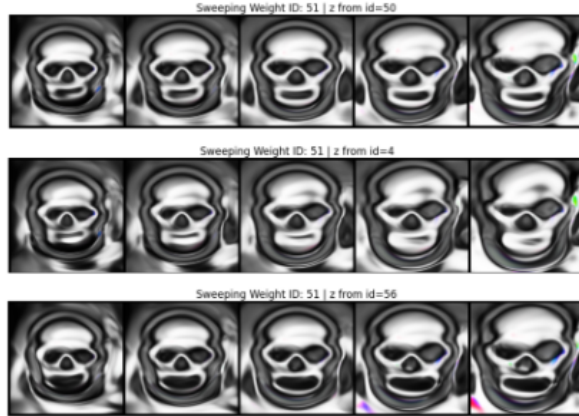


Figure 12: Sweeping weight ID 51 across three images, showing that it consistently changes the scale while keeping the unique features of each image.

When the dataset was expanded to 120 images, training became more difficult and the generated outputs appeared slightly blurrier. Nevertheless, many of the previously identified weight IDs retained similar roles. For example, weight 1909 continued to control brightness and weight 51 affected global scale. Importantly, as shown in figure 13, these effects remained meaningful even when we increased the weight variation radius to $R=2$, that would certainly cause image collapse in both the single-image setting and the 60-image dataset.

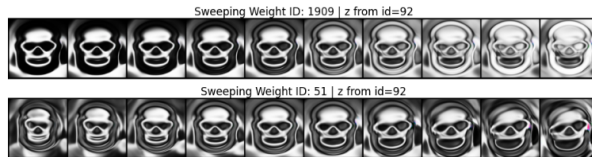


Figure 13: Weights 1909 and 51 exhibit the same semantic effects in the 120-image dataset as in the 60-image dataset and remain stable under larger weight variation ranges ($R = 2$).

Furthermore, in the larger dataset, it became significantly easier to identify additional weight IDs with interpretable and relatively localized effects. However, even in the 120-image setting, the internal representations did not approach the clean, hierarchical, and clearly factorized structure associated with UFR in open-ended evolved CPPNs. Overall, the experiments indicate that increasing dataset size introduces a mild pressure toward structured internal representations. While FER is not resolved, it becomes less catastrophic: weight variations become more stable, and certain weights acquire semi-consistent semantic roles across images. Nevertheless, scaling alone does not appear sufficient to induce a transition to true unified factored representations

3 Discussion and Future Directions

In order to understand the paper’s abstract ideas in the current dominant framework, large language models, we analysed fractured entangled representations (FER) within the broader landscape of mechanistic interpretability in transformer networks. Existing work, utilising Sparse Autoencoders (SAEs), has provided a concrete foundation for concepts such as superposition, entanglement, and feature-level decomposition. These tools make it possible to investigate internal model structure at scale and to construct mechanistic tools such as attribution graphs that reveal internal computational pathways.

However, an important limitation of current approaches is their strong dependence on the quality and assumptions of SAE-based feature extraction. Most approaches of entanglement

and fracture assume that the learned SAE features form a meaningful and sufficiently complete basis for the model’s internal representations. In practice, feature dictionaries are sensitive to several settings, such as architectural choices or sparsity constraints.

A second key assumption underlying many FER-related metrics is the Linear Representation Hypothesis, which suggests that semantically meaningful features correspond to approximately linear directions in activation space. While this assumption enables tractable measurement, recent work suggests that internal representations may exhibit richer geometric structure, including curved manifolds, feature interactions, and higher-order dependencies that are not well captured by linear projections alone.

Tools such as TRACR can be found helpful by providing transformer models with known ground-truth internal structure, allowing interpretability methods to be validated in controlled settings. Nonetheless, extracting conclusions from such synthetic constructions to large pre-trained language models remains an open challenge.

Taken together, these considerations suggest that, in order to study FER deeper in relevant architectures and define a stable and model-agnostic metric, further progress is likely required in the following directions:

- developing feature extraction methods that are less reliant on linear sparsity assumptions,
- incorporating geometric and nonlinear structure into representation analysis.

In this sense, FER should be viewed not as a measurable quantity, but as a structural hypothesis about internal representations—one that motivates ongoing methodological development rather than immediate standardization.

4 Conclusion

This study revisited the Fractured Entangled Representation (FER) hypothesis to distinguish between “Imposter Intelligence,” where models mimic output behaviors via entangled heuristics, and true structural understanding. Through a comprehensive comparative analysis of SGD-trained and evolved networks, we provided concrete evidence that the training paradigm fundamentally dictates the topology of the internal representation.

Our experiments yield three critical insights. First, we confirmed that standard SGD naturally drifts toward FER, but this tendency can be mitigated through targeted interventions; the introduction of orthogonality and sparsity regularizations forced the model toward more disentangled and interpretable weight spaces. Second, our analysis of Picbreeder lineages highlighted that Unified Factored Representations rely on the early establishment of low-dimensional structural axes, such as symmetry, which are preserved and modulated throughout the network’s depth. This stands in contrast to SGD training, which often achieves identical outputs without internalizing these foundational regularities.

Most importantly, our dataset scaling experiments challenge the assumption that more data automatically resolves representational fracture. While increasing the dataset size from 60 to 120 images improved weight stability and reduced catastrophic interference, the underlying representations remained entangled and lacked the clean modularity observed in evolutionary baselines. This suggests that FER is not merely a symptom of data scarcity, but a structural bias of the optimization process itself.

As we scale toward larger foundation models, these findings imply that benchmark performance alone is insufficient to detect “imposter” representations. Future work must move beyond linear sparsity assumptions and incorporate geometric and nonlinear analyses to better understand—and potentially correct—the fractured representations hidden within high-performing systems.

References

- https://transformer-circuits.pub/2022/toy_model/index.html
- <https://transformer-circuits.pub/2023/monosemantic-features/index.html>
- <https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html>
- <https://transformer-circuits.pub/2025/attribution-graphs/methods.html>
- <https://static1.squarespace.com/static/669758ca11526478ba0619b2/t/67ecbb6b72cdcc5d88e531743569557450/bereska-towards-measuring-superposition.pdf>
- <https://arxiv.org/pdf/2409.14507v6>
- <https://arxiv.org/pdf/2311.03658v2>
- <https://arxiv.org/pdf/2405.14860>
- <https://arxiv.org/pdf/2301.05062>